

SEMI-ANALYTIC INTEGER-ORDER MOMENTS AND SIZING OF DEFI LONG-SHORT POSITIONS UNDER HEALTH-FACTOR LIQUIDATION AND KOU JUMP-DIFFUSION RISK

We study the initial sizing of a leveraged DeFi long–short position whose solvency is governed by a protocol health factor. The position is modelled as absorbed when the health factor crosses the liquidation boundary; practical liquidation mechanics are outside the model scope. Under bivariate Kou double-exponential jump–diffusion dynamics, we reduce surviving–path payoff moments to one-dimensional killed expectations of the log collateral-to-liability ratio. An exponential change of measure absorbs the short-leg price factor, while the exponential jump structure reduces the Laplace-domain resolvent to a finite-dimensional linear system. Crucially, the construction is available for every admissible integer moment order $k \geq 1$ satisfying $k\eta_{2,+} < 1$: changing k changes the tilt and forcing term, but not the dimension of the 3×3 linear system. The framework therefore provides fast semi-analytic estimates of survival probabilities and conditional payoff moments of arbitrary admissible integer order as functions of the initial log health buffer, adapting classical first-passage machinery to the DeFi health-factor payoff setting.

Keywords: DeFi lending; leveraged long–short; health factor; Kou double-exponential jump diffusion; first-passage problem; Laplace resolvent; optimal portfolio sizing

JEL Classification: G11; G12; C65

1. Introduction

DeFi lending protocols such as AAVE and Compound enable a leveraged long–short strategy: deposit a cryptocurrency as collateral, borrow a second asset, and sell it short. The position is governed by a real-time *health factor* — a borrowing-power-adjusted ratio of collateral to liability — and is liquidated automatically and irrevocably the instant that ratio falls below one. A practitioner must choose the initial coin holdings to balance two opposing forces: a more aggressive position amplifies expected returns on surviving paths but raises the probability of forced exit. Cryptocurrency markets compound the difficulty — daily log-returns of 10–20% are common and flash crashes can exceed 50% — making fat-tailed jump dynamics essential for realistic liquidation probabilities.

This paper provides the first exact, semi-analytic solution to the optimal sizing problem under a *survival-conditioned solvent-path payoff model*: the objective is defined on the payoff realised on paths that avoid liquidation, and the liquidation boundary is treated as an absorbing barrier.

We work under the Kou double-exponential jump–diffusion framework (Kou, 2002). The solution rests on two successive dimensional reductions. First, an exponential change of measure absorbs the short-leg price factor into the path-measure density, collapsing what is nominally a bivariate barrier problem to a *univariate* first-exit problem for the log-ratio process under a tilted Lévy measure. Second,

because exponential jump sizes are eigenfunctions of the Kou generator, the Laplace-domain resolvent of the first-exit problem reduces to a 3×3 linear system — the same size for every admissible integer moment order k — that is solved in closed form. Thus the method is not restricted to the first two moments used by mean–variance sizing: it computes each integer-order raw or conditional moment for which $k\eta_{2,+} < 1$. Time-domain quantities follow by numerical Laplace inversion via the Talbot contour algorithm (Talbot, 1979). The entire optimisation therefore reduces to a scalar search over a single parameter, the *initial log health factor*, with the conditional expected payoff, its variance, and the liquidation probability all expressible as semi-analytic functions of that scalar. Unlike Monte Carlo, which suffers from sampling noise that becomes severe when few paths survive liquidation and requires fresh simulation at each candidate h_0 , the semi-analytic expressions are noise-free and differentiable in h_0 , reducing the optimization to a smooth scalar search; numerical experiments confirm a $\approx 55\times$ speed-up over Monte Carlo.

This paper sits at the intersection of the DeFi liquidation literature and the analytical first-passage literature for jump diffusions. Qin et al. (2021) and Heimbach and Huang (2024) document the incentive structure and leverage amplification of DeFi liquidations; these papers are empirical and do not address the ex-ante optimal sizing problem. Filling that gap requires tools from the first-passage literature for jump diffusions: Kou and Wang (2003) derive explicit Laplace-transform results for first-passage times under double-exponential jump diffusion, and Sepp (2004) extends this to single- and double-barrier option pricing, reducing the resolvent to a finite algebraic system — the same structural advantage exploited here. The difference is that the payoff is a barrier-killed long–short ratio, not a single-asset barrier option; the change-of-measure reduction is the key step that bridges the two settings. More distantly, Jacobs et al. (1999) advocate joint optimisation of long and short legs, while Dang and Forsyth (2016) study dynamic mean–variance control under jump risk; our setting is static and the binding constraint is an endogenous first-passage event rather than a variance bound.

The paper makes three contributions to the sizing of leveraged DeFi positions. First, it establishes the initial log health factor h_0 as the natural scalar sizing variable: once a unit-initial-value constraint is imposed, the two-dimensional portfolio problem collapses to a scalar health-buffer decision, with the raw killed moments $\mathbb{E}[\Pi_T^k]$ for every admissible integer order $k \geq 1$ satisfying $k\eta_{2,+} < 1$, and the liquidation probability $p_{\text{liq}}(h_0)$, all explicit in h_0 alone. Second, it delivers those quantities semi-analytically — noise-free, differentiable in h_0 , and $\approx 55\times$ faster than Monte Carlo. For each admissible integer k , the calculation requires a new tilt and Laplace inversion but retains the same 3×3 resolvent system; its algebraic dimension does not grow with k . This makes gradient-based optimisation of any criterion of the form $\Phi(\mathbb{E}[\Pi_T | \tau > T], \dots, \mathbb{E}[\Pi_T^K | \tau > T], p_{\text{liq}})$ practical for any finite admissible integer K . Third, it characterises how the optimal buffer trades conditional upside against liquidation risk, providing a principled alternative to the ad-hoc collateral ratios used in practice. The framework abstracts away liquidation execution, dynamic collateral management, stochastic borrow rates, and oracle delays; it is a liquidation-buffer evaluation engine for ex-ante position sizing, not a complete trading system.

The remainder of the paper is organised as follows. Section 2 introduces the position, health factor, payoff, and the one-dimensional reduction to a scalar optimisation. Section 3 specifies the asset dynamics under the Kou model; Section 3.2 introduces the tilted-measure machinery. Section 4 shows how the change of measure decouples the short-leg factor from the barrier problem, reducing each raw killed moment $\mathbb{E}[\Pi_T^k]$ to a one-dimensional killed expectation for every integer $k \geq 1$ satisfying $k\eta_{2,+} < 1$; dividing by $p_{\text{surv}}(h_0)$ yields the conditional moments entering the general objective (5). Section 5 derives the respective resolvents. Section 6 presents numerical results, comparing the Laplace–resolvent method against Monte Carlo simulation across a range of initial log health factor values and verifying the analytical no-shorting limit. Section 7 concludes and discusses the practical interpretation of the results. Appendix A describes the empirical characteristic function estimator used to calibrate the bivariate Kou parameters from observed bivariate log-returns. Appendix B collects the proofs.

2. Problem Setup

We consider a leveraged long–short portfolio in which an investor holds a long position in asset S_1 and a short position in asset S_2 , financed so that the net initial value of the book equals one unit of account. The strategy is subject to a risk-management rule that forces liquidation if the long leg falls sufficiently relative to the short leg, reflecting margin requirements or stop-loss constraints common in practice. The objective is to choose the initial coin holdings so as to optimise a performance criterion defined on the terminal payoff, conditional on surviving the liquidation threshold.

2.1. Surviving-path terminal wealth

Let $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ be a filtered probability space. Two strictly positive asset prices S_1, S_2 evolve as

$$S_{i,t} = S_{i,0} e^{X_{i,t}}, \quad X_{i,0} = 0, \quad i = 1, 2,$$

where X_1, X_2 are log-price processes to be specified in Section 3. The investor holds $n_1 > 0$ units of asset S_1 (long leg) and $n_2 > 0$ units of asset S_2 (short leg).

Let

$$V_{1,t} := n_1 S_{1,t}, \quad V_{2,t} := n_2 S_{2,t}$$

denote the *collateral value* (long leg) and the *borrow value* (outstanding liability) at time t . Fix $b \in (0, 1)$, the *liquidation threshold* of asset S_1 (the Aave v3 parameter of the same name; distinct from the maximum LTV at origination, which is a separate and typically lower protocol constant): the protocol liquidates the position the first time the borrow value exceeds the b -adjusted collateral,

$$b V_{1,t} \geq V_{2,t}.$$

The *health factor* summarises this margin condition as a single ratio,

$$H_t := \frac{b V_{1,t}}{V_{2,t}} = \frac{b n_1 S_{1,t}}{n_2 S_{2,t}}, \quad h_t := \log H_t, \quad (1)$$

where h_t is the *log health factor*. The solvency condition is $H_t \geq 1$ ($h_t \geq 0$); breaching it is equivalent to the borrow value exceeding the b -adjusted collateral. Liquidation is triggered at the first such instant:

$$\tau := \inf\{t \geq 0 : H_t < 1\} = \inf\{t \geq 0 : h_t < 0\}.$$

A larger b permits a larger borrow-to-collateral ratio $V_2/V_1 \leq b$ before liquidation, giving the position more room to absorb adverse price moves; a smaller b is more conservative, triggering earlier forced exit and limiting the protocol's risk of bad debt.

Since

$$h_t = \log\left(\frac{b n_1 S_{1,0}}{n_2 S_{2,0}}\right) + (X_{1,t} - X_{2,t}),$$

define the initial log health factor and the log-ratio process by

$$h_0 := \log\left(\frac{b n_1 S_{1,0}}{n_2 S_{2,0}}\right), \quad Z_t := X_{1,t} - X_{2,t},$$

so that $h_t = h_0 + Z_t$ and the liquidation time is equivalently

$$\tau = \inf\{t \geq 0 : Z_t < -h_0\}.$$

The one-dimensional process Z thus inherits all the relevant risk: the position survives if and only if Z stays above the absorbing level $-h_0$. The parameter $h_0 = \log H_0$ is therefore the *initial log-cushion* — the log-distance of the initial health factor from the liquidation boundary $H = 1$. A larger h_0 means the position starts further from forced exit; $h_0 \rightarrow 0^+$ means the position starts almost at the boundary with negligible cushion.

The *surviving-path terminal wealth* at horizon T is

$$\Pi_T := (n_1 S_{1,T} - n_2 S_{2,T}) \mathbf{1}_{\{\tau > T\}}.$$

On $\{\tau > T\}$ the health factor satisfies $H_T \geq 1$, so $b n_1 S_{1,T} \geq n_2 S_{2,T}$; since $b \in (0, 1)$ this forces $n_1 S_{1,T} > n_2 S_{2,T}$, meaning the position carries strictly positive terminal equity on every surviving path. No positive-part truncation is needed. The indicator $\mathbf{1}_{\{\tau > T\}}$ assigns zero terminal wealth to liquidated paths; the remark below explains why zero is used as a conservative proxy for the residual recovery value. Since the initial net book value is normalised to one unit of account (Section 2.2), net profit on a surviving path is $R_T := \Pi_T - 1$, but all moments in this paper are stated in terms of Π_T directly. This paper characterises the distribution of Π_T on the event $\{\tau > T\}$ and the probability of that event, with the boundary $H_t = 1$ treated as an absorbing state; the liquidation mechanics are outside the model scope.

Remark 1 (Liquidation mechanics). When H_t hits one, a DeFi lending protocol does not simply extinguish the position: it initiates a protocol-specific liquidation procedure executed by third-party keepers. In AAVE v3, a keeper may close up to 50% of the debt (100% if the health factor falls below a secondary threshold), claiming a *liquidation bonus* of 5–15% of the seized collateral as compensation; Compound uses a fixed close factor with a similar bonus structure (Qin et al., 2021; Heimbach and Huang, 2024).

The execution price received by the keeper — and hence the residual recovery value for the borrower — is itself uncertain. Liquidations in DeFi are settled on-chain via decentralised exchanges: in stressed markets, the available liquidity may be insufficient to absorb a large collateral sale without significant slippage, compressing the borrower’s recovery further (Qin et al., 2021). The mechanics therefore depend on the specific protocol version, the pool liquidity at the moment of liquidation, the gas price environment, and the speed of keeper response — none of which are captured by a single-barrier first-passage model.

For these reasons we assign zero payoff to liquidated paths: $\Pi_T := 0$ on $\{\tau \leq T\}$. This is a *conservative proxy* — it understates the expected residual value whenever the borrower retains a fractional recovery, and overstates the loss relative to protocols that execute partial liquidations. Its practical justification is two-fold. First, the borrower typically has no control over the execution details once the boundary is breached; from an ex-ante sizing perspective the liquidation payoff is a zero-mean random variable with uncertain sign, and setting it to zero is the risk-neutral midpoint. Second, the penalty term δp_{liq} in the objective of (6) provides a separate lever to price liquidation risk at whatever level the investor deems appropriate, without requiring an explicit model of the post-liquidation cash flow. Assigning zero to liquidated paths is therefore *conservative* — it never overstates residual value — and *practical*: it avoids an unverifiable recovery model while keeping the payoff specification tractable.

Remark 2 (Dynamic collateral management). A practitioner who monitors the health factor in continuous time could top up collateral or partially unwind the short leg to steer H_t away from the liquidation boundary. Such a strategy would replace the absorbing barrier with a reflecting or controlled boundary and the static sizing problem with a stochastic control problem. We do not pursue this extension for two reasons. First, on-chain collateral top-ups incur gas costs and are subject to block-confirmation delays; during a flash crash the position may be liquidated before a remedial transaction is confirmed, so the absorbing-barrier model represents the practically relevant worst case. Second, the paper’s central question is the ex-ante buffer decision — how much initial cushion to hold given that dynamic intervention may be unavailable — and a static framework isolates that question cleanly.

Taken together, the two modelling choices — zero recovery at liquidation and no dynamic intervention — define a *conservative floor* for position sizing: any position that is viable under these assumptions remains viable whenever protocol outcomes are more favourable.

Using the definition of H_T , one obtains the following factorization.

Lemma 2.1 (Payoff factorization).

$$\Pi_T = \frac{n_2}{b} S_{2,0} e^{X_{2,T}} (e^{h_0 + Z_T} - b) \mathbf{1}_{\{\tau > T\}}. \quad (2)$$

The factor $(e^{h_0 + Z_T} - b)$ is strictly positive on $\{\tau > T\}$ because $H_T = e^{h_0 + Z_T} \geq 1 > b$, consistent with the definition of Π_T .

This separates the payoff into a numeraire factor driven by S_2 , and a barrier-killed option on the log-ratio Z with moneyness controlled by h_0 .

2.2. Optimization setup

The two coin holdings (n_1, n_2) represent two degrees of freedom. A normalisation reduces this to a single scalar, and the structure of the payoff then identifies h_0 as the natural optimisation variable.

We impose the unit-initial-value constraint

$$n_1 S_{1,0} - n_2 S_{2,0} = 1,$$

which fixes the net book value at inception. Setting $x := n_2 S_{2,0}$, one has $n_1 S_{1,0} = 1 + x$ and $e^{h_0} = b(1 + x)/x$. Solving for x ,

$$x = \frac{b}{e^{h_0} - b},$$

which yields the holding map

$$n_2(h_0) = \frac{b}{S_{2,0}(e^{h_0} - b)}, \quad n_1(h_0) = \frac{e^{h_0}}{S_{1,0}(e^{h_0} - b)}. \quad (3)$$

Positivity of both holdings requires $e^{h_0} > b$, i.e.

$$h_0 > \ln b.$$

The initial gross leverage — long-leg value per unit of equity — follows directly from (3):

$$L_0 := n_1(h_0) S_{1,0} = \frac{e^{h_0}}{e^{h_0} - b}. \quad (4)$$

L_0 is strictly decreasing in h_0 : as $h_0 \downarrow \ln b$, leverage $L_0 \uparrow \infty$; as $h_0 \rightarrow \infty$, leverage $L_0 \downarrow 1$ (the short leg vanishes and the position becomes unlevered). Equivalently, the borrow-to-equity ratio is $L_0 - 1 = b/(e^{h_0} - b)$, which also collapses the initial health factor back to $H_0 = e^{h_0}$ via $H_0 = b L_0 / (L_0 - 1)$. Choosing h_0 is therefore equivalent to choosing the initial leverage.

Since every quantity of interest is determined by h_0 through (3) and the killed process (Z, τ) , the portfolio problem reduces to a scalar optimisation

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0).$$

The optimal holdings are recovered by substituting h_0^* back into (3); we call h_0^* the *optimal initial health buffer*.

The framework supports any performance criterion of the form

$$\Phi(h_0) = f(\mathbb{E}[\Pi_T | \tau > T], \dots, \mathbb{E}[\Pi_T^K | \tau > T], p_{\text{liq}}(h_0)), \quad (5)$$

where $\mathbb{E}[\Pi_T^k | \tau > T] = \mathbb{E}[\Pi_T^k] / p_{\text{surv}}(h_0)$ are the conditional moments obtained from the raw killed moments $\mathbb{E}[\Pi_T^k]$ delivered in Section 4, $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$, and f is any smooth function. The admissible range of moment orders is $k = 1, \dots, K$ with $K\eta_{2,+} < 1$, a bound determined by the integrability of $e^{kX_{2,T}}$ under the exponential tilt (Proposition 4.1). Here K is not fixed by the method: any positive integer K satisfying the integrability bound may be selected, and the same construction is applied separately at each $k \in \{1, \dots, K\}$. This class covers the classical mean–variance criterion (Markowitz, 1952), which uses $\mathbb{E}[\Pi_T | \tau > T]$ and $\mathbb{E}[\Pi_T^2 | \tau > T]$; higher-moment objectives that penalise skewness or excess kurtosis, which use $\mathbb{E}[\Pi_T | \tau > T], \dots, \mathbb{E}[\Pi_T^4 | \tau > T]$; the conditional Sharpe ratio; and polynomial approximations to expected utility of arbitrary order. Dang and Forsyth (2016) and Shen and Zou (2022) adopt mean–variance criteria in related jump-diffusion settings; the distinguishing feature here is that $p_{\text{liq}}(h_0)$ is an explicit, separately computed building block rather than an implicit risk charge.

For the numerical illustration in Section 6 we adopt the penalised conditional mean–variance criterion

$$\Phi(h_0) := \mathbb{E}[\Pi_T | \tau > T] - \frac{\alpha}{2} \text{Var}(\Pi_T | \tau > T) - \delta p_{\text{liq}}(h_0), \quad (6)$$

where $\alpha \geq 0$ is a risk-aversion coefficient and $\delta \geq 0$ penalises forced liquidation. The optimal h_0^* balances leverage against downside protection: a lower h_0 implies higher leverage and a larger conditional upside but a higher liquidation probability; a higher h_0 provides more cushion at the cost of a smaller conditional payoff. We do not refer to h_0^* as “optimal leverage” because leverage is a derived quantity (4); the investor chooses the health buffer and leverage follows.

The remainder of the paper is devoted to computing the building blocks of (5): the raw killed moments $\mathbb{E}[\Pi_T^k]$ for every admissible integer $k \geq 1$ satisfying $k\eta_{2,+} < 1$, and the survival probability $p_{\text{surv}}(h_0)$, each as an explicit semi-analytic function of h_0 via the Laplace–resolvent method of Section 5. The conditional moments $\mathbb{E}[\Pi_T^k | \tau > T]$ and liquidation probability $p_{\text{liq}}(h_0)$ then follow immediately.

3. Asset Dynamics

3.1. Bivariate Kou model

We assume under $\mathbb{P}$ that each log-price satisfies the SDE

$$dX_{i,t} = \mu_i^X dt + \sigma_i dB_{i,t} + d\left(\sum_{n=1}^{N_{i,t}} J_i^{(n)}\right), \quad i = 1, 2, \quad (7)$$

where B_1, B_2 are correlated standard Brownian motions with $d\langle B_1, B_2 \rangle_t = \rho dt$, N_i is a Poisson process with intensity λ_i independent of (B_1, B_2) , and the jump sizes $J_i^{(n)}$ are i.i.d. with the Kou double-exponential law. The jump density, characteristic function, and moment generating function are

$$f_i(j) = \frac{p_i}{\eta_{i,+}} e^{-j/\eta_{i,+}} \mathbf{1}_{\{j>0\}} + \frac{1-p_i}{\eta_{i,-}} e^{j/\eta_{i,-}} \mathbf{1}_{\{j<0\}}, \quad \varphi_{J_i}(u) = \frac{p_i}{1-iu\eta_{i,+}} + \frac{1-p_i}{1+iu\eta_{i,-}}, \quad (8)$$

$$M_i(s) = \frac{p_i}{1-s\eta_{i,+}} + \frac{1-p_i}{1+s\eta_{i,-}}, \quad s \in \left(-\frac{1}{\eta_{i,-}}, \frac{1}{\eta_{i,+}}\right). \quad (9)$$

Here $\eta_{i,+}$ and $\eta_{i,-}$ are the *means* of the positive and negative jump sizes (not exponential rates); the corresponding rates are $1/\eta_{i,+}$ and $1/\eta_{i,-}$. The constraint $\eta_{i,+} < 1$ is required for the price-jump compensator χ_i defined below to be finite.

The parameter μ_i is the *continuously compounded expected asset-price growth rate*, i.e. $\mathbb{E}_{\mathbb{P}}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$. Define the price-jump compensator

$$\chi_i := M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1] = \frac{p_i}{1 - \eta_{i,+}} + \frac{1 - p_i}{1 + \eta_{i,-}} - 1. \quad (10)$$

Since $\mathbb{E}_{\mathbb{P}}[e^{X_{i,t}}] = \exp(t[\mu_i^X + \frac{1}{2}\sigma_i^2 + \lambda_i\chi_i])$, matching to $e^{\mu_i t}$ gives the log-price drift coefficient

$$\mu_i^X := \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i\chi_i. \quad (11)$$

All analytical and numerical objects in the sequel use μ_i^X .

The bivariate process (X_1, X_2) is a Lévy process with joint characteristic exponent

$$\begin{aligned} \Psi(u, v) := & i(\mu_1^X u + \mu_2^X v) - \frac{1}{2}(\sigma_1^2 u^2 + \sigma_2^2 v^2 + 2\rho\sigma_1\sigma_2 uv) \\ & + \lambda_1(\varphi_{J_1}(u) - 1) + \lambda_2(\varphi_{J_2}(v) - 1), \end{aligned} \quad (12)$$

so that $\mathbb{E}_{\mathbb{P}}[e^{i(uX_{1,t} + vX_{2,t})}] = e^{t\Psi(u,v)}$.

Table 1 collects all model parameters and their roles.

3.2. Tilted Dynamics

For each integer $k \geq 0$, define the k -tilted measure $\mathbb{P}^{(k)}$ by

$$\left. \frac{d\mathbb{P}^{(k)}}{d\mathbb{P}} \right|_{\mathcal{F}_t} = \frac{e^{kX_{2,t}}}{\mathbb{E}_{\mathbb{P}}[e^{kX_{2,t}}]} = \exp(kX_{2,t} - t\Psi(0, -ik)), \quad k\eta_{2,+} < 1.$$

Lemma 3.1 (Lévy exponent under $\mathbb{P}^{(k)}$). *Under $\mathbb{P}^{(k)}$, the log-ratio $Z = X_1 - X_2$ is again a Lévy process with exponent*

$$\psi_Z^{(k)}(s) = \mu_Z^{(k)}s + \frac{1}{2}\sigma_Z^2 s^2 + \lambda_1(M_1(s) - 1) + \lambda_2(M_2(k-s) - M_2(k)), \quad (13)$$

where

$$\sigma_Z^2 := \sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2 \geq 0, \quad \mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2).$$

In particular, the generator satisfies

$$\mathcal{L}^{(k)} e^{sz} = \psi_Z^{(k)}(s) e^{sz}. \quad (14)$$

▷ *Proof, p. 23*

This eigenrelation is the key identity used in Section 5 to build the resolvent equation.

Remark 3 (Superscript (k) convention). A superscript (k) indicates evaluation under $\mathbb{P}^{(k)}$. The untilded case $k = 0$ recovers the original dynamics of Z under $\mathbb{P}$. For $k \geq 1$, the k -th moment $\mathbb{E}[\Pi_T^k]$ is computed under $\mathbb{P}^{(k)}$; the constraint $k\eta_{2,+} < 1$ ensures the tilted measure and the characteristic root structure remain well-defined. Throughout, k is an integer. There is no restriction to $k = 1$ or $k = 2$: every positive integer satisfying the integrability condition is covered.

For the Kou structure, the tilt transforms the jump rates of Z into the k -dependent phase rates

$$r_{1,+}^{(k)} := \frac{1}{\eta_{1,+}}, \quad r_{1,-}^{(k)} := \frac{1}{\eta_{1,-}}, \quad r_{2,+}^{(k)} := \frac{1}{\eta_{2,-}} + k, \quad r_{2,-}^{(k)} := \frac{1}{\eta_{2,+}} - k.$$

Because $Z = X_1 - X_2$, positive jumps of X_2 appear as downward jumps of Z : the tilt by e^{kX_2} changes the downward phase rate from $1/\eta_{2,+}$ to $1/\eta_{2,+} - k$, which is positive precisely when $k\eta_{2,+} < 1$.

Symbol	Asset	Economic meaning / role	Constraint
μ_i	$i = 1, 2$	Continuously compounded expected price growth rate: $\mathbb{E}[S_{i,t}/S_{i,0}] = e^{\mu_i t}$	
σ_i	$i = 1, 2$	Diffusion volatility of log-price X_i	$\sigma_i > 0$
ρ	—	Correlation between the two Brownian drivers	$ \rho < 1$
λ_i	$i = 1, 2$	Jump arrival intensity (expected jumps per unit time)	$\lambda_i > 0$
p_i	$i = 1, 2$	Probability that a jump in X_i is upward	$p_i \in (0, 1)$
$\eta_{i,+}$	$i = 1, 2$	Mean size of upward jumps in X_i	$0 < \eta_{i,+} < 1$
$\eta_{i,-}$	$i = 1, 2$	Mean size of downward jumps in X_i	$\eta_{i,-} > 0$
χ_i	$i = 1, 2$	Price-jump compensator $M_i(1) - 1 = \mathbb{E}[e^{J_i} - 1]$; enters μ_i^X via (11)	
μ_i^X	$i = 1, 2$	Actual drift coefficient in the log-price SDE (7)	
$\Psi(u, v)$	—	Joint characteristic exponent; $e^{T\Psi(0, -ik)} = \mathbb{E}[e^{kX_{2,T}}]$ appears in the moment formula	
$S_{i,0}$	$i = 1, 2$	Initial price of asset i	$S_{i,0} > 0$
b	—	Liquidation threshold: maximum borrow-to-collateral ratio V_2/V_1 before forced liquidation (Aave v3 liquidation threshold; distinct from maximum LTV)	$b \in (0, 1)$
T	—	Investment horizon	$T > 0$
n_i	$i = 1, 2$	Number of coins held: $n_1 > 0$ long in S_1 , $n_2 > 0$ short in S_2	$n_i > 0$
H_t	—	Health factor: $b n_1 S_{1,t} / (n_2 S_{2,t})$; liquidated when $H_t < 1$	
h_t	—	Log health factor: $h_t = \log H_t = h_0 + Z_t$	
h_0	—	Initial log health factor; the optimisation variable; determines n_i via (3)	$h_0 > \log b$
τ	—	Liquidation time: $\tau = \inf\{t \geq 0 : H_t < 1\}$	
Π_T	—	Surviving-path terminal wealth: $(n_1 S_{1,T} - n_2 S_{2,T}) \mathbf{1}_{\{\tau > T\}}$	

Table 1: Model parameters and derived quantities. The first block contains freely chosen Kou model inputs; the second block contains derived model quantities; the third block contains portfolio and contract inputs; the fourth block contains derived portfolio quantities.

Definition 3.1 (Phase operators). For $j = 1, 2$, the *phase operators* associated with the tilted jump kernels of Z under $\mathbb{P}^{(k)}$ are

$$K_{j,+}^{(k)}[f](z) := \int_0^\infty f(z+y) r_{j,+}^{(k)} e^{-r_{j,+}^{(k)} y} dy, \quad K_{j,-}^{(k)}[f](z) := \int_{-\infty}^0 f(z+y) r_{j,-}^{(k)} e^{r_{j,-}^{(k)} y} dy.$$

This construction originates in [Kou and Wang \(2003\)](#) for first-passage problems under the double-exponential model, and is used by [Sepp \(2004\)](#) to reduce double-barrier pricing to a finite linear system. For any γ with $\gamma < r_{j,+}^{(k)}$ and $\gamma > -r_{j,-}^{(k)}$, evaluating the integrals directly gives the eigenfunction identity

$$K_{j,+}^{(k)}[e^\gamma](z) = \frac{r_{j,+}^{(k)}}{r_{j,+}^{(k)} - \gamma} e^{\gamma z}, \quad K_{j,-}^{(k)}[e^\gamma](z) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma} e^{\gamma z}. \quad (15)$$

Each phase operator maps an exponential to a scalar multiple of the same exponential, which is the key algebraic identity exploited in Section 5 to convert the resolvent PIDE into a finite-dimensional linear system.

Table 2 collects the quantities introduced in this subsection.

Symbol	Definition	Role
$\mathbb{P}^{(k)}$	$d\mathbb{P}^{(k)} / d\mathbb{P} \propto e^{kX_{2,t}}$	Measure under which moment reduction is carried out
k	Non-negative integer, $k\eta_{2,+} < 1$	Tilting order; $k \geq 1$ for moments, $k = 0$ for survival probability
$\psi_Z^{(k)}(s)$	Lévy exponent of Z under $\mathbb{P}^{(k)}$	Characteristic root equation $\psi_Z^{(k)}(\gamma) = q$ governs the resolvent
σ_Z^2	$\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2$	Variance of the Brownian part of Z ; appears in $\psi_Z^{(k)}$
$\mu_Z^{(k)}$	$(\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho\sigma_1\sigma_2)$	Drift of Z under $\mathbb{P}^{(k)}$; appears in $\psi_Z^{(k)}$
$\mathcal{L}^{(k)}$	Generator of Z under $\mathbb{P}^{(k)}$	$\mathcal{L}^{(k)}e^{sz} = \psi_Z^{(k)}(s)e^{sz}$; used in the resolvent PIDE
$r_{j,\pm}^{(k)}$	Phase rates (see above)	Exponential rates of the jump components of Z under $\mathbb{P}^{(k)}$
$K_{j,\pm}^{(k)}$	Phase operators (see above)	Convert jump integrals to scalars on exponential ansatz

Table 2: Tilted-dynamics quantities. Subscript $j = 1$ refers to the contribution from X_1 jumps; $j = 2$ to that from X_2 jumps (sign-reversed because $Z = X_1 - X_2$).

4. Moments via Measure Change

Fix any integer $k \geq 1$ satisfying $k\eta_{2,+} < 1$. Computing $\mathbb{E}_{\mathbb{P}}[\Pi_T^k]$ directly under $\mathbb{P}$ is intractable: the payoff couples the joint process (X_1, X_2) to the barrier event $\{\tau > T\}$ in a way that is neither separable nor analytically tractable. An exponential change of measure resolves this by tilting the path measure by $e^{kX_{2,T}}$, which absorbs the short-leg factor into the density and leaves a purely one-dimensional killed expectation for the log-ratio $Z = X_1 - X_2$ under the k -tilted measure $\mathbb{P}^{(k)}$. Since Z remains a Lévy process under $\mathbb{P}^{(k)}$, no information about the joint dynamics is lost: the tilted characteristic exponent encodes the full marginal law of Z . The k -th moment then factorises as the known scalar $e^{T\Psi(0, -ik)} = \mathbb{E}_{\mathbb{P}}[e^{kX_{2,T}}]$ times the barrier-killed expectation $m_k(T; h_0)$ under $\mathbb{P}^{(k)}$, which is computed via the Laplace–resolvent of Section 5.

The argument below is order-generic. It is applied independently at each desired integer k , and the resulting resolvent always has a 3×3 barrier system. Higher integer orders enlarge only the finite forcing sum from $j = 0, \dots, k$; they do not enlarge the linear system.

Define

$$g_k(z; h_0) := (e^{h_0+z} - b)^k.$$

Since the survival event $\{\tau > T\}$ implies $Z_T > -h_0$, we have $e^{h_0+Z_T} > 1 > b$ on $\{\tau > T\}$, so $g_k(Z_T; h_0) = (e^{h_0+Z_T} - b)^k > 0$ and is a polynomial in e^{Z_T} . From (2),

$$\begin{aligned} \Pi_T^k &= \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{kX_{2,T}} g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}. \\ \mathbb{E}_{\mathbb{P}}[\Pi_T^k] &= \left(\frac{n_2}{b}\right)^k S_{2,0}^k e^{T\Psi(0, -ik)} \mathbb{E}_{\mathbb{P}^{(k)}}[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}]. \end{aligned} \quad (16)$$

Proposition 4.1 (Every admissible integer-order moment). *Let $k \geq 1$ be an integer satisfying $k\eta_{2,+} < 1$. Define*

$$m_k(T; h_0) := \mathbb{E}_{\mathbb{P}^{(k)}}[g_k(Z_T; h_0) \mathbf{1}_{\{\tau > T\}}].$$

Then

$$\mathbb{E}[\Pi_T^k] := \mathbb{E}_{\mathbb{P}}[\Pi_T^k] = \frac{e^{T\Psi(0, -ik)}}{(e^{h_0} - b)^k} m_k(T; h_0),$$

where $m_k(T; h_0) = U^{(k)}(T, 0; h_0)$ is computed via the Laplace–resolvent method of Section 5. Consequently, the method computes $\mathbb{E}[\Pi_T^k]$ for every integer

$$k \in \mathcal{K} := \{n \in \mathbb{N} : n\eta_{2,+} < 1\}.$$

Let $p_{\text{surv}}(h_0) := \mathbb{P}(\tau > T)$ and $p_{\text{liq}}(h_0) := 1 - p_{\text{surv}}(h_0)$. Since Π_T already contains the survival indicator $\mathbf{1}_{\{\tau > T\}}$, the conditional k -th moment on surviving paths is

$$\mathbb{E}[\Pi_T^k \mid \tau > T] = \frac{\mathbb{E}[\Pi_T^k]}{p_{\text{surv}}(h_0)}, \quad k \geq 1, \quad k\eta_{2,+} < 1.$$

In particular,

$$\mathbb{E}[\Pi_T \mid \tau > T] = \frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)}, \quad \text{Var}(\Pi_T \mid \tau > T) = \frac{\mathbb{E}[\Pi_T^2]}{p_{\text{surv}}(h_0)} - \left(\frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)} \right)^2.$$

The mean–variance–liquidation objective is therefore

$$\Phi(h_0) = \frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)} - \frac{\alpha}{2} \left[\frac{\mathbb{E}[\Pi_T^2]}{p_{\text{surv}}(h_0)} - \left(\frac{\mathbb{E}[\Pi_T]}{p_{\text{surv}}(h_0)} \right)^2 \right] - \delta p_{\text{liq}}(h_0), \quad (17)$$

and the portfolio optimization reduces to the one-dimensional problem

$$\max_{h_0 \in (\ln b, \infty)} \Phi(h_0),$$

after which the optimal coin holdings are recovered from (3).

Remark 4 (No-shorting limit). As $h_0 \rightarrow \infty$ the holding map gives $n_2 S_{2,0} \rightarrow 0$ and $n_1 S_{1,0} \rightarrow 1$, so liquidation becomes impossible and the portfolio converges to a unit long position in asset 1. For every $k \geq 1$,

$$\lim_{h_0 \rightarrow \infty} \mathbb{E}[\Pi_T^k] = \mathbb{E}[e^{kX_{1,T}}] = e^{T\Psi(-ik,0)},$$

which provides a useful diagnostic: the formula for $\mathbb{E}[\Pi_T^k]$ must satisfy this limit exactly. The $k = 1$ case gives $e^{\mu_1 T}$ by the drift normalisation (11).

The remainder of the analysis is devoted to computing $m_k(T; h_0)$ for every admissible integer $k \geq 1$ and $p_{\text{surv}}(h_0)$ via the Laplace–resolvent construction.

5. Resolvent Equations for Moments and Survival Probability

Proofs of all results in Sections 3.2–5 are collected in Appendix B.

With the moment problem reduced to a one-dimensional killed expectation under $\mathbb{P}^{(k)}$, the function $U(t, z) = \mathbb{E}_z^{(k)}[g_k(Z_t; h_0) \mathbf{1}_{\{\tau > t\}}]$ satisfies a PIDE in (t, z) . Taking the Laplace transform in t converts this to a stationary integro-differential equation in z alone, with the Laplace parameter acting as a discount rate. The Kou structure then reduces it further: because exponentials are eigenfunctions of the generator, every integro-differential term acts algebraically on an exponential ansatz, and the resolvent collapses to a finite-dimensional linear system. Three boundary conditions at the absorbing barrier close the 3×3 system. Importantly, this system remains 3×3 for every admissible integer k ; increasing the moment order changes its coefficients and the forcing polynomial, not its dimension. The survival probability follows by the same architecture at zero tilt ($k = 0$). This section derives both resolvents in closed form.

5.1. Killed Moment Resolvent

Fix an integer $k \geq 1$ with $k\eta_{2,+} < 1$. Define the killed moment function

$$U(t, z) := \mathbb{E}_z^{(k)}[g_k(Z_t; h_0) \mathbf{1}_{\{\tau(h_0) > t\}}], \quad z > -h_0,$$

with $U(t, z) = 0$ for $z \leq -h_0$. This satisfies the PIDE

$$\partial_t U = \mathcal{L}^{(k)} U, \quad z > -h_0,$$

with initial condition $U(0, z) = g_k(z; h_0)$. Define the Laplace transform

$$\widehat{U}(q, z) := \int_0^\infty e^{-qt} U(t, z) dt, \quad q > 0.$$

Applying it to both sides of the PIDE proceeds in two steps.

Left-hand side. Integrate by parts in t :

$$\int_0^\infty e^{-qt} \partial_t U(t, z) dt = \left[e^{-qt} U(t, z) \right]_0^\infty + q \int_0^\infty e^{-qt} U(t, z) dt = -U(0, z) + q \widehat{U}(q, z),$$

where the boundary term at $t = \infty$ vanishes by integrability.

Right-hand side. Since $\mathcal{L}^{(k)}$ acts only in z , it commutes with the t -integral:

$$\int_0^\infty e^{-qt} \mathcal{L}^{(k)} U(t, z) dt = \mathcal{L}^{(k)} \widehat{U}(q, z).$$

Equating and substituting the initial condition $U(0, z) = g_k(z; h_0)$ gives the resolvent equation

$$(q - \mathcal{L}^{(k)}) \widehat{U}(q, z) = g_k(z; h_0), \quad z > -h_0, \quad (18)$$

with $\widehat{U}(q, z) = 0$ for $z \leq -h_0$.

Assumption 5.1 (Generic six-root regime). Fix an integer $k \geq 0$ satisfying $k\eta_{2,+} < 1$, and let $q > 0$. Assume:

(i) $\sigma_Z > 0$;

(ii) the characteristic equation $\psi_Z^{(k)}(\gamma) = q$ has six simple roots $\gamma_1(q), \dots, \gamma_6(q)$, ordered so that

$$\operatorname{Re}(\gamma_{1,2,3}) > 0, \quad \operatorname{Re}(\gamma_{4,5,6}) < 0;$$

(iii) $q \neq \psi_Z^{(k)}(j)$ for $j = 0, \dots, k$.

Since $b \in (0, 1)$ and $z > -h_0$ together imply $e^{h_0+z} > 1 > b$, the payoff kernel expands as a polynomial in e^z :

$$g_k(z; h_0) = \sum_{j=0}^k c_{k,j}(h_0) e^{jz}, \quad c_{k,j}(h_0) := \binom{k}{j} (-1)^{k-j} b^{k-j} e^{jh_0}. \quad (19)$$

Lemma 5.1 (Particular solution). *Under Assumption 5.1(iii),*

$$\widehat{U}_{\text{part}}(q, z; h_0) = \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} e^{jz} \quad (20)$$

is a particular solution to (18).

▷ *Proof, p. 23*

Lemma 5.2 (Homogeneous solution). *Under Assumption 5.1(i)–(ii), the general solution to $(q - \mathcal{L}^{(k)})u = 0$ on $z > -h_0$ that is admissible at $+\infty$ is*

$$\widehat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^3 B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)(z+h_0)}, \quad (21)$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants.

▷ Proof, p. 24

The general solution on $z > -h_0$ is then

$$\widehat{U}(q, z; h_0) = \widehat{U}_{\text{part}}(q, z; h_0) + \widehat{U}_{\text{hom}}(q, z; h_0).$$

Proposition 5.3 (Barrier resolvent). *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(k)} \neq r_{2,-}^{(k)}, \quad (22)$$

the boundary conditions at $z = -h_0^+$ uniquely determine $\mathbf{B}^{(k)}(q; h_0) := (B_1^{(k)}, B_2^{(k)}, B_3^{(k)})^\top$ via the linear system

$$\mathbf{M}_{\text{bar}}^{(k)}(q) \mathbf{B}^{(k)}(q; h_0) = -\mathbf{b}^{(k)}(q; h_0),$$

where the 3×3 barrier matrix is

$$\mathbf{M}_{\text{bar}}^{(k)}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_4(q)} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_5(q)} & \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + \gamma_6(q)} \\ \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_4(q)} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_5(q)} & \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + \gamma_6(q)} \end{pmatrix}$$

and the forcing vector is

$$\mathbf{b}^{(k)}(q; h_0) = \begin{pmatrix} \widehat{U}_{\text{part}}(q, -h_0; h_0) \\ K_{1,-}^{(k)} \left[\widehat{U}_{\text{part}}(q, \cdot; h_0) \right](-h_0) \\ K_{2,-}^{(k)} \left[\widehat{U}_{\text{part}}(q, \cdot; h_0) \right](-h_0) \end{pmatrix} = \begin{pmatrix} \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} e^{-jh_0} \\ \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} \frac{r_{1,-}^{(k)}}{r_{1,-}^{(k)} + j} e^{-jh_0} \\ \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} \frac{r_{2,-}^{(k)}}{r_{2,-}^{(k)} + j} e^{-jh_0} \end{pmatrix}.$$

By Lemma 5.4 below, $\mathbf{M}_{\text{bar}}^{(k)}(q)$ is invertible, giving

$$\mathbf{B}^{(k)}(q; h_0) = -(\mathbf{M}_{\text{bar}}^{(k)}(q))^{-1} \mathbf{b}^{(k)}(q; h_0).$$

▷ Proof, p. 24

Lemma 5.4 (Invertibility of $\mathbf{M}_{\text{bar}}^{(k)}(q)$). *Under Assumption 5.1 and condition (22), the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}(q)$ of Proposition 5.3 is invertible.*

Evaluating $\widehat{U}$ at the initial state $Z_0 = 0$ gives

$$\widehat{U}(q, 0; h_0) = \sum_{j=0}^k \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)} + \sum_{m=1}^3 B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)h_0}. \quad (23)$$

The killed moment is then recovered by Laplace inversion:

$$m_k(T; h_0) = \mathcal{L}_{q \rightarrow T}^{-1} \left[\widehat{U}(q, 0; h_0) \right].$$

5.2. Survival Probability

The same resolvent architecture yields the survival probability. Define

$$p_{\text{surv}}(t, z; h_0) := \mathbb{P}_z(\tau(h_0) > t), \quad p_{\text{liq}}(t, z; h_0) := 1 - p_{\text{surv}}(t, z; h_0).$$

Let $\mathcal{L}^{(0)}$ denote the generator of Z under the untitled measure, with exponent $\psi_Z^{(0)}$, and define

$$\widehat{U}_{\text{surv}}(q, z; h_0) := \int_0^\infty e^{-qt} p_{\text{surv}}(t, z; h_0) dt.$$

The problem is

$$(q - \mathcal{L}^{(0)})\widehat{U}_{\text{surv}}(q, z; h_0) = 1, \quad z > -h_0,$$

with

$$\widehat{U}_{\text{surv}}(q, z; h_0) = 0, \quad z \leq -h_0.$$

The domain is the half-line $(-h_0, \infty)$, the boundary is absorbing at $z = -h_0$, and there is no interface because the source is constant.

Proposition 5.5 (Survival resolvent and liquidation probability). *Under Assumption 5.1 and the nondegeneracy condition*

$$r_{1,-}^{(0)} \neq r_{2,-}^{(0)}, \quad \text{i.e.,} \quad \eta_{1,-} \neq \eta_{2,+}, \quad (24)$$

let $\gamma_4^{(0)}(q), \gamma_5^{(0)}(q), \gamma_6^{(0)}(q)$ be the three roots of $\psi_Z^{(0)}(\gamma) = q$ with negative real part. Then

$$\widehat{U}_{\text{surv}}(q, z; h_0) = \frac{1}{q} + C_1(q; h_0) e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2(q; h_0) e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3(q; h_0) e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where the coefficients $\mathbf{C}(q; h_0) := (C_1, C_2, C_3)^\top$ satisfy the linear system

$$\mathbf{M}_{\text{surv}}(q) \mathbf{C}(q; h_0) = - \begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

with the 3×3 survival matrix

$$\mathbf{M}_{\text{surv}}(q) = \begin{pmatrix} 1 & 1 & 1 \\ \frac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_4^{(0)}(q)} & \frac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_5^{(0)}(q)} & \frac{r_{1,-}^{(0)}}{r_{1,-}^{(0)} + \gamma_6^{(0)}(q)} \\ \frac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_4^{(0)}(q)} & \frac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_5^{(0)}(q)} & \frac{r_{2,-}^{(0)}}{r_{2,-}^{(0)} + \gamma_6^{(0)}(q)} \end{pmatrix}.$$

By Lemma 5.6 below, $\mathbf{M}_{\text{surv}}(q)$ is invertible, giving

$$\mathbf{C}(q; h_0) = -(\mathbf{M}_{\text{surv}}(q))^{-1} \begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix}.$$

At the initial state $Z_0 = 0$,

$$\widehat{U}_{\text{surv}}(q, 0; h_0) = \frac{1}{q} + C_1(q; h_0)e^{\gamma_4^{(0)}(q)h_0} + C_2(q; h_0)e^{\gamma_5^{(0)}(q)h_0} + C_3(q; h_0)e^{\gamma_6^{(0)}(q)h_0}.$$

Hence

$$p_{\text{surv}}(T, 0; h_0) = \mathcal{L}_{q \rightarrow T}^{-1} \left[\widehat{U}_{\text{surv}}(q, 0; h_0) \right], \quad p_{\text{liq}}(T, 0; h_0) = 1 - p_{\text{surv}}(T, 0; h_0).$$

▷ Proof, p. 25

Lemma 5.6 (Invertibility of $\mathbf{M}_{\text{surv}}(q)$). *Under Assumption 5.1 and condition (24), the survival matrix $\mathbf{M}_{\text{surv}}(q)$ of Proposition 5.5 is invertible.*

▷ Proof, p. 26

6. Numerical Results

We validate the Laplace–resolvent method against Monte Carlo (MC) simulation for $k = 0, 1, 2, 3, 4$ and verify the analytical no-shorting limit as $h_0 \rightarrow \infty$. These orders illustrate rather than limit the method: the same implementation applies to every admissible integer k , which under the calibrated parameters includes $k = 1, \dots, 79$. Time-domain quantities are recovered via the Talbot deformed-contour algorithm (Talbot, 1979). No Bromwich shift needs to be chosen by hand.

The Kou model and contract parameters are given in Table 3. Parameters are calibrated by ECF matching to 540 four-hour WBTC (asset 1) and WETH (asset 2) log-returns on AAVE v3 Ethereum after subtracting an exponentially weighted return mean; the drifts μ_i are adjusted for AAVE supply/borrow APRs (see caption). The AAVE liquidation threshold $b = 0.78$ and maximum origination LTV 0.73 impose a minimum feasible log-buffer $h_0^{\min} = \log(0.78/0.73) \approx 0.066$. WBTC and WETH have distinct downward phase rates $r_{1,-}^{(0)} = 1/\eta_{1,-} \neq r_{2,+}^{(0)} = 1/\eta_{2,+}$, so the barrier matrix $\mathbf{M}_{\text{bar}}^{(k)}$ is non-singular for every admissible integer k (here the calibrated parameters permit integer orders through $k = 79$). Because the first moment is weakly identified in high-frequency ECF matching, the fitted μ_i values in Table 3 should be read as model price-growth normalizers rather than directional forecasts; user views can replace them in downstream moment and liquidation calculations.

For each initial health factor $H_0 = e^{h_0}$, with $h_0 \in \{0.07, 0.10, 0.25, 0.50, 0.75, 1.00, 1.50, 2.00\}$, we report the survival probability p_{surv} and the following conditional statistics of $\Pi_T \mid \{\tau > T\}$, where $\mu_k := \mathbb{E}[\Pi_T^k \mid \tau > T]$:

$$\begin{aligned} \text{mean} &:= \mu_1, & \text{variance} &:= \mu_2 - \mu_1^2, \\ \text{skewness} &:= \frac{\mu_3 - 3\mu_1\mu_2 + 2\mu_1^3}{(\mu_2 - \mu_1^2)^{3/2}}, & \text{ex. kurt} &:= \frac{\mu_4 - 4\mu_1\mu_3 + 6\mu_1^2\mu_2 - 3\mu_1^4}{(\mu_2 - \mu_1^2)^2} - 3. \end{aligned} \quad (25)$$

As $h_0 \rightarrow \infty$ the short holding $n_2 \rightarrow 0$ and the portfolio converges to a unit long position in asset 1 (Remark 4), so all conditional statistics converge to those of $e^{X_{1,T}}$ under $\mathbb{P}$:

$$\lim_{h_0 \rightarrow \infty} \mu_k = \mathbb{E}_{\mathbb{P}} \left[e^{kX_{1,T}} \right] = \exp(T\Psi(-ik, 0)).$$

Table 4 reports all statistics together with these analytical limits.

Group	Parameter	WBTC	WETH
Kou	ECF price-growth normalizer μ_i	0.487	1.200
	Log-price drift μ_i^X	-0.305	0.257
	Model log-return drift $\mathbb{E}[dX_i]/dt$	0.425	1.111
	Empirical de-meaned log drift	0.352	0.606
	Diffusion volatility σ_i	0.241	0.307
	Jump intensity λ_i	372.95	588.95
	Upward-jump probability p_i	0.508	0.357
	Mean upward-jump size $\eta_{i,+}$	0.0111	0.0125
	Mean downward-jump size $\eta_{i,-}$	0.00743	0.00469
Correlation / contract	Brownian correlation ρ	0.858	
	Collateral factor b	0.78	
Portfolio	Initial price $S_{i,0}$	75,060.57	2,060.33
	Horizon T	1/12 (one month)	

Table 3: Kou model and contract parameters calibrated by ECF matching to four-hour WBTC/WETH log-returns on AAVE v3 Ethereum (2026-02-26 – 2026-05-27), after EWM de-meaning with span 30. Drifts adjusted for AAVE average APRs: $\mu_1 = -0.0056\% \text{ yr}^{-1}$ (WBTC supply), $\mu_2 = 2.7855\% \text{ yr}^{-1}$ (WETH borrow). Jump intensities are estimated under the broad calibration bound $\lambda_{\max} = 5000$. AAVE liquidation threshold $b = 0.78$; maximum origination LTV 0.73 gives $h_0^{\min} \approx 0.066$. Derived quantities: $\chi_1 + \frac{1}{2}\sigma_1^2 \approx 0.793$, $\chi_2 + \frac{1}{2}\sigma_2^2 \approx 0.943$, $\mu_1^X \approx -0.305$, $\mu_2^X \approx 0.257$. The empirical de-meaned price-growth normalizers are 0.432 (WBTC) and 0.755 (WETH), so the fitted WETH μ_2 is a weakly identified ECF normalizer rather than a robust return forecast. Initial prices are the last aligned AAVE sample, at block 25,189,091.

6.1. Parameter sensitivity

Figure 1 shows how p_{surv} (left column) and the conditional mean $\mathbb{E}[\Pi_T \mid \tau > T]$ (right column) vary with H_0 as each parameter is swept in turn. For the collateral factor b (row e–f), survival probability is independent of b at fixed H_0 — only the payoff kernel changes — so the left panel shows the initial leverage $L_0 = H_0/(H_0 - b)$ instead.

Three patterns dominate. (i) *Volatility* is the strongest driver: at $0.3\times$ scale, almost all paths survive for $H_0 \gtrsim 1.35$; at $2\times$ scale, the same cushion yields survival below 40%. (ii) *Correlation* acts through the effective variance of $Z = X_1 - X_2$: the calibrated benchmark already sits at $\rho = 0.858$, where the spread variance is small; moving to lower ρ sharply depresses the survival curve, while negligible effect on the conditional mean confirms that correlation works purely through barrier-crossing risk. (iii) *Jump intensity* and *horizon* compress survival monotonically, while increasing the conditional mean at low H_0 through selective survivor effects. For b , initial leverage $L_0 = H_0/(H_0 - b)$ reaches $\approx 6.3\times$ at the AAVE minimum $H_0^{\min} \approx 1.068$ for $b = 0.90$, explaining the strong sensitivity of the conditional mean to b near the minimum buffer.

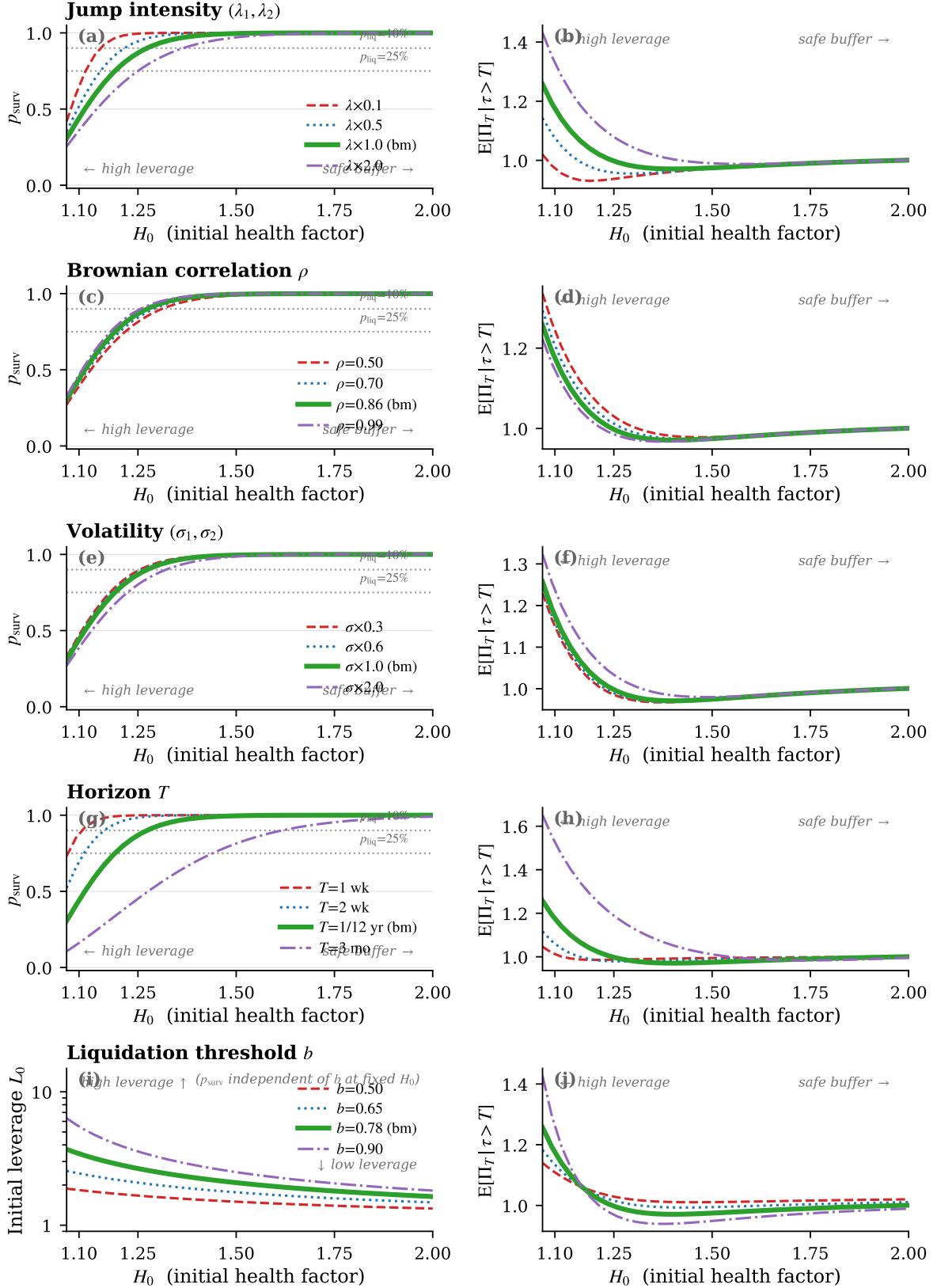


Figure 1: Left column: survival probability p_{surv} (rows a–d) and initial leverage L_0 (row e, log scale) versus H_0 . Right column: conditional mean $E[\Pi_T | \tau > T]$. Each row varies one parameter; all others fixed at Table 3. Encoding: colour and line style identify parameter values (solid/green = benchmark); the heavier line is the baseline. Laplace–resolvent, $H_0 \in [1.068, 2.000]$, 40 points per curve.

h_0	H_0	p_{surv}		Mean	Variance	Skewness	Ex. Kurt.
		Lap.	MC				
0.07	1.07	0.3051	0.3163	1.2581	0.0753	0.8767	1.0996
0.10	1.11	0.4615	0.4750	1.1633	0.0672	0.8525	1.0083
0.25	1.28	0.9129	0.9156	0.9847	0.0545	0.5676	0.3668
0.50	1.65	0.9996	0.9995	0.9845	0.0350	0.3132	0.3565
0.75	2.12	1.0000	1.0000	1.0043	0.0236	0.3791	0.4093
1.00	2.72	1.0000	1.0000	1.0158	0.0186	0.4158	0.4234
1.50	4.48	1.0000	1.0000	1.0280	0.0145	0.4318	0.4187
2.00	7.39	1.0000	1.0000	1.0339	0.0130	0.4263	0.4017
∞	∞	1.0000	—	1.0414	0.0114	0.4052	0.3612

Table 4: Laplace–resolvent conditional statistics and the no-shorting analytical limit for WBTC/WETH calibrated parameters (Table 3). Mean, variance, skewness, and excess kurtosis are conditional on survival ($\tau > T$) and computed via the resolvent method (Laplace); MC benchmark uses 10^5 Euler–Poisson paths (252 steps, seed 42). The smallest tabulated $h_0 = 0.07$ is the AAVE-feasible minimum ($h_0^{\min} \approx 0.066$, $H_0^{\min} \approx 1.068$); at this value Laplace gives $p_{\text{surv}} = 0.305$ versus MC 0.316 — the 1.1% gap reflects discrete-time MC missing intra-period liquidation events at high jump intensity ($\lambda_1 \approx 373$, $\lambda_2 \approx 589$), not path scarcity ($\approx 32,000$ surviving MC paths). The bottom row gives the analytical limits as $h_0 \rightarrow \infty$; Laplace at $h_0 = 100$ matches these values to relative error $< 10^{-6}$, confirming internal consistency of the resolvent construction, Talbot inversion, and the drift normalisation $\mu_i^X = \mu_i - \frac{1}{2}\sigma_i^2 - \lambda_i\chi_i$.

h_0	H_0	Laplace – MC					CPU time (s)	
		Δp_{surv}	ΔMean	$\Delta \text{Var.}$	$\Delta \text{Skew.}$	$\Delta \text{Kurt.}$	Lap.	MC
0.07	1.07	−0.0112	0.0042	−0.0019	−0.0233	−0.1959	0.10	50.37
0.10	1.11	−0.0136	0.0042	−0.0013	−0.0230	−0.1499	0.10	49.15
0.25	1.28	−0.0026	−0.0001	−0.0007	−0.0054	−0.0563	0.10	48.90
0.50	1.65	0.0001	−0.0005	−0.0003	−0.0110	−0.0347	0.10	49.11
0.75	2.12	0.0000	−0.0002	−0.0002	−0.0107	−0.0382	0.10	49.49
1.00	2.72	0.0000	−0.0002	−0.0002	−0.0123	−0.0378	0.10	49.46
1.50	4.48	0.0000	−0.0001	−0.0001	−0.0144	−0.0369	0.10	49.94
2.00	7.39	0.0000	0.0000	−0.0001	−0.0152	−0.0362	0.10	49.76

Table 5: Laplace minus MC differences ($\Delta = \text{Lap.} - \text{MC}$) and wall-clock computation time. The persistent $\Delta p_{\text{surv}} < 0$ reflects the discrete-time Euler–Poisson scheme (252 steps yr^{-1}) missing intra-period liquidation events; the effect is strongest near $H_0^{\min} \approx 1.068$ where paths frequently approach the barrier. Higher-moment differences shrink as H_0 increases and more paths survive into the interior. The Laplace method covers five Talbot inversions ($k = 0, \dots, 4$); the $\approx 490\times$ speed-up over MC is uniform across all tabulated values.

7. Conclusion and Discussion

This paper develops a semi-analytic method for sizing a leveraged DeFi long–short position subject to health-factor liquidation. The initial log health factor h_0 reduces the sizing problem to one dimension, while an exponential change of measure reduces each payoff moment to a killed expectation of the log-ratio process. Under Kou jump–diffusion dynamics, the corresponding Laplace resolvent is obtained from a 3×3 linear system.

A central feature is that the construction applies to every admissible integer order $k \geq 1$ satisfying $k\eta_{2,+} < 1$. Higher-order moments require a different tilt and forcing polynomial, but do not increase the dimension of the resolvent system. The framework therefore supports mean–variance sizing as well as criteria involving skewness, kurtosis, higher-moment penalties, or polynomial utility approximations. The numerical results show close agreement with Monte Carlo, while the semi-analytic method is substantially faster and free of simulation noise.

For practical use, an investor can first choose a maximum liquidation probability and invert $p_{\text{liq}}(H_0)$ to obtain a minimum health factor. Within the resulting feasible set, the investor then maximises the preferred moment-based objective. The optimal leverage follows from the selected health factor through $L_0 = H_0/(H_0 - b)$. Figure 2 summarises this tradeoff between conditional payoff, variance, and liquidation probability.

The model deliberately abstracts from oracle delays, liquidation execution and recovery, dynamic collateral adjustments, and stochastic borrowing rates. It should therefore be read as an ex-ante liquidation-buffer and sizing tool rather than a complete trading or execution model.

Further work may incorporate liquidation recovery, dynamic collateral management, and stochastic funding costs.

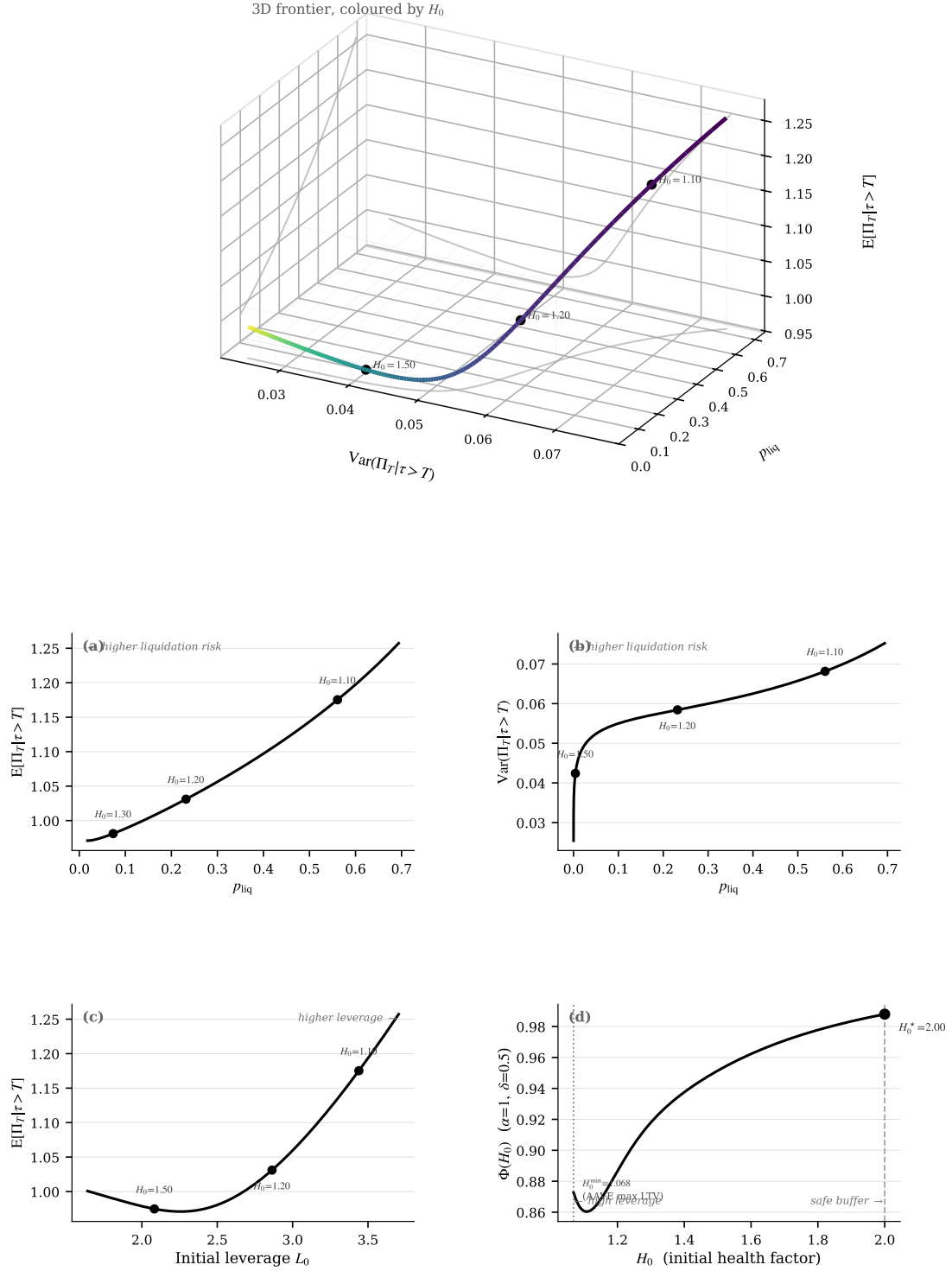


Figure 2: Mean–variance–liquidity frontier of $\Pi_T | \{\tau > T\}$ for the benchmark parametrisation. Axes: $x = \text{Var}(\Pi_T | \tau > T)$, $y = p_\ell = 1 - p_{\text{surv}}$, $z = E[\Pi_T | \tau > T]$ (vertical). The curve is parametrised by $H_0 \in [1.068, 2.000]$; decreasing H_0 moves the curve toward higher variance and higher p_ℓ , while the conditional mean becomes U-shaped near the liquidation boundary. Grey projections on the three coordinate planes recover the pairwise two-dimensional tradeoffs; panel (a) plots the increasing efficient branch of the mean–liquidity projection. Script `jobs/frontier_analysis.py`.

Appendix A: Empirical Calibration

A.1. Estimator

The bivariate Kou model admits the closed-form Lévy–Khintchine exponent $\Psi(u, v)$ derived in Section 3, so no numerical density inversion is required to compute the model-implied characteristic function

$$\phi_\theta(u, v; \Delta) = \exp(\Delta \Psi_\theta(u, v)), \quad (26)$$

where Δ is the observation interval. Given N observed bivariate log-return pairs $(r_{1,n}, r_{2,n})_{n=1}^N$, the empirical characteristic function is

$$\hat{\phi}_N(u, v) = \frac{1}{N} \sum_{n=1}^N \exp(i(ur_{1,n} + vr_{2,n})). \quad (27)$$

We estimate θ by minimising the weighted squared distance between (27) and (26) over a discrete frequency grid $\mathcal{G}$:

$$\hat{\theta} = \arg \min_{\theta \in \Theta} Q_N(\theta), \quad Q_N(\theta) = \sum_{(u,v) \in \mathcal{G}} w(u, v) |\hat{\phi}_N(u, v) - \phi_\theta(u, v; \Delta)|^2. \quad (28)$$

This is the empirical-characteristic-function approach of Singleton (2001), which is particularly natural for affine and Lévy-based asset-pricing models whose characteristic functions are known in closed form. Feuerverger and McDunnough (1981) establish that ECF estimators over a rich grid can attain maximum-likelihood efficiency. The procedure avoids evaluating the bivariate transition density, which would require a two-dimensional numerical Fourier inversion inside every optimizer call. For background on DEJD likelihood estimation in the univariate case see Ramezani and Zeng (2007).

A.2. Frequency grid

The parameter vector θ has 13 free components; see Table 1. Identification is most efficient when the frequency grid $\mathcal{G}$ probes the directions that are informative for different parameter groups. We use three groups:

- $\mathcal{G}_1$ (*marginal directions*): $\{(u, 0) : u \in \mathcal{U}\} \cup \{(0, v) : v \in \mathcal{V}\}$. These directions identify the marginal Kou parameters of each asset independently.
- $\mathcal{G}_2$ (*joint directions*): $\{(u, v) : u, v \in \mathcal{U}_{\text{joint}}\}$. Cross-frequency pairs identify the Brownian correlation ρ .
- $\mathcal{G}_Z$ (*spread direction*): $\{(s, -s) : s \in \mathcal{U}_Z\}$. Along this direction $ur_1 + vr_2 = s(r_1 - r_2)$, so the estimator directly matches the characteristic function of the log-ratio process $Z = X_1 - X_2$ that governs DeFi liquidation. Including $\mathcal{G}_Z$ explicitly ties the calibration to the object that drives barrier crossing.

The full grid is $\mathcal{G} = \mathcal{G}_1 \cup \mathcal{G}_2 \cup \mathcal{G}_Z$. Default frequency nodes are $\mathcal{U} = \{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0, 4.0, 5.0\}$, $\mathcal{U}_{\text{joint}} = \{0.5, 1.0, 2.0\}$, $\mathcal{U}_Z = \{0.25, 0.5, 0.75, 1.0, 1.5, 2.0, 3.0\}$, giving $M = 34$ moment conditions for 13 parameters. Weights use Gaussian damping $w(u, v) = \exp(-a(u^2 + v^2))$ with $a = 1$, which down-weights high frequencies sensitive to microstructure noise.

A.3. Reparameterisation and initialisation

Unconstrained reparameterisation. Constraints in Table 1 are enforced automatically via the change of variables

$$\sigma_i = e^{a_i}, \quad \lambda_i = e^{b_i}, \quad p_i = \text{sigmoid}(c_i), \quad \eta_{i,+} = \text{sigmoid}(d_i), \quad \eta_{i,-} = e^{e_i}, \quad \rho = \tanh(f), \quad (29)$$

where $\text{sigmoid}(x) = (1 + e^{-x})^{-1}$. The unconstrained vector $\tau = (\mu_1, a_1, b_1, c_1, d_1, e_1, \mu_2, \dots, f)$ is optimized with box bounds in this transformed space. For asset 2 the code uses a scaled sigmoid,

$$\eta_{2,+} = \eta_{2,+}^{\max} \text{sigmoid}(d_2), \quad \eta_{2,+}^{\max} = (1 - \varepsilon)/K,$$

so that the moment-admissibility condition $K\eta_{2,+} < 1$ is enforced directly. For asset 1, the sigmoid bound $\eta_{1,+} < 1$ keeps the price-jump compensator finite.

Threshold initialiser. The characteristic-function objective is non-linear and non-convex, so the optimisation is initialised by a threshold-based decomposition following the logic of Mancini (2009) for discretely observed jump-diffusions. Let $m_i = \text{median}(r_{i,n})$ and let s_i be the IQR robust scale. For a frequency-aware threshold constant $c \in [3, 5]$, define candidate jump observations

$$\mathcal{J}_i = \{n : |r_{i,n} - m_i| > cs_i\}, \quad \mathcal{C}_i = \{1, \dots, N\} \setminus \mathcal{J}_i.$$

Initial volatility and correlation values use the continuous subset $\mathcal{C}_i$, while the jump-size means are anchored by peak-over-threshold excess means and the intensities are corrected by a variance/fourth-cumulant estimate. Schematically:

$$\begin{aligned} p_i^{(0)} &= \frac{|\mathcal{J}_{i,+}|}{|\mathcal{J}_i|} \quad \text{clamped to the admissible probability range,} \\ \eta_{i,+}^{(0)}, \eta_{i,-}^{(0)} &= \text{POT mean-excess estimates,} \\ \lambda_i^{(0)} &= \text{two-moment estimate from variance and fourth cumulant,} \\ \sigma_i^{(0)} &= \text{sd}(r_{i,n} : n \in \mathcal{C}_i) / \sqrt{\Delta}, & \rho^{(0)} &= \text{corr}(r_1, r_2)_{n \in \mathcal{C}_1 \cap \mathcal{C}_2}. \end{aligned}$$

The initializer is stored in the same drift convention as the final result: $\mu_i^{(0)}$ is an annualized expected price-growth drift. It is initialized from the sample log-return drift with a damped mean-jump correction,

$$\mu_i^{(0)} = \bar{r}_i / \Delta - 0.25 \lambda_i^{(0)} \left[p_i^{(0)} \eta_{i,+}^{(0)} - (1 - p_i^{(0)}) \eta_{i,-}^{(0)} \right],$$

and the corresponding log-process coefficient $\mu_i^{X,(0)}$ is obtained from (11). The threshold filter is not the final estimator. It is a warm start that places the optimizer in a sensible region of parameter space; see the remark in Mancini (2009).

Multi-start cloud. Because $Q_N(\tau)$ may have local minima, the optimizer is run from a cloud of n_{start} starting points (default 20). Each point perturbs the threshold initialiser by applying a multiplicative factor $\exp(U)$, $U \sim \text{Uniform}(-0.4, 0.4)$, to each scale parameter, and an additive shift $\varepsilon \sim \text{Uniform}(-0.15, 0.15)$ to p_i and ρ . The estimate returned is the unconstrained L-BFGS-B minimiser with the lowest objective value across all starts.

A.4. Identifiability and model comparison

The ECF objective $Q_N(\theta)$ is non-convex. Jump-diffusion models possess approximate ridges along directions that preserve the total jump variance $\lambda_i(p_i\eta_{i,+}^2 + (1-p_i)\eta_{i,-}^2)$ (Ramezani and Zeng, 2007). The diffusion volatilities σ_i and the Brownian correlation ρ are well-identified from short samples; jump-size and intensity parameters carry larger finite-sample uncertainty unless the observation window contains many jump events. For hourly cryptocurrency data the expected number of jump observations per trading year is $\lambda_i \times 8760$, which is typically in the hundreds; daily data with $\lambda_i = 2$ yields only ≈ 16 identifiable jumps per year.

For model comparison, evaluate $Q_N(\hat{\theta})$ across nested models (GBM, Merton, bivariate Kou) together with liquidation-relevant diagnostics: empirical versus model Z-tail quantiles, empirical versus model barrier-crossing frequencies, and simulated versus observed drawdowns of Z . These are more directly informative for the sizing application than standard likelihood-based criteria.

The implementation is in `optimal_long_short/calibration.py`; the demo script `jobs/calibrate_kou.py` shows parameter recovery on synthetic data and produces Figure 3.

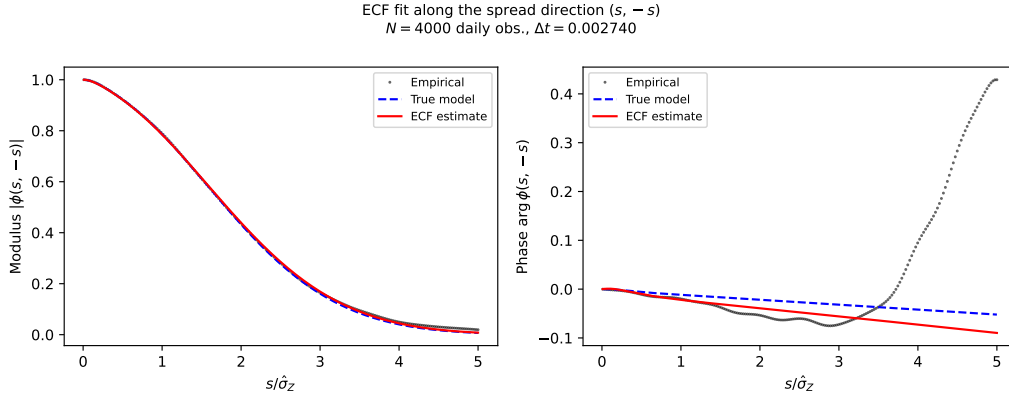


Figure 3: ECF fit along the spread direction ($s, -s$) for $N = 4,000$ synthetic daily log-return observations. Left: modulus $|\phi(s, -s)|$; right: phase $\arg \phi(s, -s)$. Dots: empirical CF $\hat{\phi}_N$; dashed: true model CF; solid red: ECF estimate. The estimator matches the empirical CF accurately across both modulus and phase, confirming that $Q_N(\hat{\theta}) \approx 0$. Script `jobs/calibrate_kou.py`.

Appendix B: Proofs

Proof of Lemma 3.1. Step 1: Laplace transform under $\mathbb{P}^{(k)}$. Because $Z_t = X_{1,t} - X_{2,t}$ and the density on $\mathcal{F}_t$ equals $e^{kX_{2,t} - t\Psi(0, -ik)}$,

$$\mathbb{E}_{\mathbb{P}^{(k)}}[e^{sZ_t}] = e^{-t\Psi(0, -ik)} \mathbb{E}_{\mathbb{P}}[e^{sX_{1,t} + (k-s)X_{2,t}}] = e^{t[\Psi(-is, -i(k-s)) - \Psi(0, -ik)]}, \quad (30)$$

where we used $\mathbb{E}_{\mathbb{P}}[e^{\alpha X_{1,t} + \beta X_{2,t}}] = e^{t\Psi(-i\alpha, -i\beta)}$. Since the Radon–Nikodým density is exponential-linear in $(X_{1,t}, X_{2,t})$, Girsanov’s theorem for Lévy processes guarantees that Z remains a Lévy process under $\mathbb{P}^{(k)}$ with exponent

$$\psi_Z^{(k)}(s) = \Psi(-is, -i(k-s)) - \Psi(0, -ik). \quad (31)$$

Step 2: Expand and simplify. Substituting (12) with $(u, v) = (-is, -i(k-s))$, and noting $u^2 = -s^2$, $v^2 = -(k-s)^2$, $uv = -s(k-s)$, and $\phi_{J_i}(-i \cdot) = M_i(\cdot)$:

$$\begin{aligned} \Psi(-is, -i(k-s)) &= \mu_1^X s + \mu_2^X (k-s) + \frac{1}{2} \sigma_1^2 s^2 + \frac{1}{2} \sigma_2^2 (k-s)^2 + \rho \sigma_1 \sigma_2 s(k-s) \\ &\quad + \lambda_1 (M_1(s) - 1) + \lambda_2 (M_2(k-s) - 1). \end{aligned} \quad (32)$$

Setting $s = 0$ in (32) gives $\Psi(0, -ik) = \mu_2^X k + \frac{1}{2} \sigma_2^2 k^2 + \lambda_2 (M_2(k) - 1)$. Subtracting and collecting terms by type:

Drift. $\mu_1^X s + \mu_2^X (k-s) - \mu_2^X k = (\mu_1^X - \mu_2^X)s$.

Diffusion. Expanding $(k-s)^2 = k^2 - 2ks + s^2$ and $(k-s) = k-s$:

$$\begin{aligned} &\frac{1}{2} \sigma_1^2 s^2 + \frac{1}{2} \sigma_2^2 (k-s)^2 + \rho \sigma_1 \sigma_2 s(k-s) - \frac{1}{2} \sigma_2^2 k^2 \\ &= \frac{1}{2} (\sigma_1^2 + \sigma_2^2 - 2\rho \sigma_1 \sigma_2) s^2 - k(\sigma_2^2 - \rho \sigma_1 \sigma_2) s = \frac{1}{2} \sigma_Z^2 s^2 - k(\sigma_2^2 - \rho \sigma_1 \sigma_2) s. \end{aligned}$$

Jumps. $\lambda_1 (M_1(s) - 1) + \lambda_2 (M_2(k-s) - 1) - \lambda_2 (M_2(k) - 1) = \lambda_1 (M_1(s) - 1) + \lambda_2 (M_2(k-s) - M_2(k))$.

Combining, the drift coefficient is $\mu_Z^{(k)} := (\mu_1^X - \mu_2^X) - k(\sigma_2^2 - \rho \sigma_1 \sigma_2)$, which establishes (13). The eigenrelation (14) follows because for any Lévy process with exponent ψ , the generator satisfies $\mathcal{L}e^{sz} = \psi(s)e^{sz}$, as can be seen by differentiating $\mathbb{E}[e^{sZ_t}] = e^{t\Psi(s)}$ with respect to t at $t = 0$. $\square$

Proof of Lemma 5.1. Substitute the forcing expansion into the right-hand side of (18):

$$(q - \mathcal{L}^{(k)})\widehat{U}_{\text{part}} = \sum_{j=0}^k c_{k,j}(h_0) e^{jz}.$$

Since the sum is finite and $\mathcal{L}^{(k)}$ is linear, it suffices to find, for each j , a function v_j satisfying

$$(q - \mathcal{L}^{(k)})v_j(z) = c_{k,j}(h_0) e^{jz}.$$

Propose the exponential ansatz $v_j(z) = A_j e^{jz}$. By the eigenrelation (14),

$$\mathcal{L}^{(k)}[A_j e^{jz}] = A_j \psi_Z^{(k)}(j) e^{jz},$$

so

$$(q - \mathcal{L}^{(k)})[A_j e^{jz}] = A_j (q - \psi_Z^{(k)}(j)) e^{jz}.$$

Matching to $c_{k,j}(h_0) e^{jz}$ and using Assumption 5.1(iii) to ensure $q - \psi_Z^{(k)}(j) \neq 0$ gives

$$A_j = \frac{c_{k,j}(h_0)}{q - \psi_Z^{(k)}(j)}.$$

Summing over $j = 0, \dots, k$ yields (20). $\square$

Proof of Lemma 5.2. Step 1: exponential ansatz. Seek solutions of the form $u(z) = e^{\gamma z}$. By the eigenrelation (14),

$$\mathcal{L}^{(k)} e^{\gamma z} = \psi_Z^{(k)}(\gamma) e^{\gamma z},$$

so $(q - \mathcal{L}^{(k)})e^{\gamma z} = (q - \psi_Z^{(k)}(\gamma))e^{\gamma z}$. This vanishes if and only if γ satisfies the characteristic equation

$$\psi_Z^{(k)}(\gamma) = q.$$

Step 2: root count. The exponent $\psi_Z^{(k)}$ is the sum of a quadratic term $\frac{1}{2}\sigma_Z^2\gamma^2$ (non-degenerate by Assumption 5.1(i)) and rational terms from the Kou jumps. Clearing denominators yields a degree-six polynomial in γ , so there are exactly six roots $\gamma_1, \dots, \gamma_6$ (counted with multiplicity). Under Assumption 5.1(ii) all six are simple and split as $\text{Re}(\gamma_{1,2,3}) > 0$ and $\text{Re}(\gamma_{4,5,6}) < 0$.

Step 3: admissibility. On the half-line $z > -h_0$, the solution must remain bounded as $z \rightarrow +\infty$. Since $e^{\gamma z} \rightarrow \infty$ whenever $\text{Re}(\gamma) > 0$, the roots $\gamma_1, \gamma_2, \gamma_3$ are inadmissible. Only $\gamma_4, \gamma_5, \gamma_6$ are retained.

Step 4: centering and general solution. Writing $e^{\gamma_{m+3}z} = e^{\gamma_{m+3}(z+h_0)} \cdot e^{-\gamma_{m+3}h_0}$ and absorbing the constant factor into the free coefficient, the general admissible homogeneous solution is

$$\hat{U}_{\text{hom}}(q, z; h_0) = \sum_{m=1}^3 B_m^{(k)}(q; h_0) e^{\gamma_{m+3}(q)(z+h_0)},$$

where $B_1^{(k)}, B_2^{(k)}, B_3^{(k)}$ are free constants to be determined by the boundary conditions. $\square$

Proof of Proposition 5.3. Step 1: boundary conditions from killing. The killing condition $\hat{U}(q, z; h_0) = 0$ for $z \leq -h_0$ imposes, at $z = -h_0^+$,

- (i) $\hat{U}(q, -h_0^+; h_0) = 0;$
- (ii) $K_{j,-}^{(k)}[\hat{U}(q, \cdot; h_0)](-h_0^+) = 0, \quad j = 1, 2.$

Condition (i) is continuity at the barrier. Conditions (ii) arise because $K_{j,-}^{(k)}[f](z) = \int_{-\infty}^0 f(z+y) r_{j,-}^{(k)} e^{r_{j,-}^{(k)}y} dy$ integrates f over landing points $z+y < z$; for $z = -h_0^+$ every landing point satisfies $z+y < -h_0$, where $\hat{U} = 0$, so the integral vanishes.

Step 2: substitute the full solution. Write $\hat{U} = \hat{U}_{\text{part}} + \hat{U}_{\text{hom}}$ and insert into each condition. Since the conditions are linear, each splits as

$$[\text{condition on } \hat{U}_{\text{hom}}] = -[\text{condition on } \hat{U}_{\text{part}}].$$

Step 3: condition (i). Evaluating at $z = -h_0$ and using $e^{\gamma_{m+3}(q) \cdot 0} = 1$,

$$\hat{U}_{\text{hom}}(q, -h_0; h_0) = \sum_{m=1}^3 B_m^{(k)} = -\hat{U}_{\text{part}}(q, -h_0; h_0),$$

which is the first row of the system.

Step 4: conditions (ii). Apply $K_{j,-}^{(k)}$ to $\hat{U}_{\text{hom}}$. By the phase-operator formula (15),

$$K_{j,-}^{(k)}[e^{\gamma_{m+3}(\cdot+h_0)}](-h_0) = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma_{m+3}(q)} e^{\gamma_{m+3}(q) \cdot 0} = \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma_{m+3}(q)},$$

giving row $j+1$ of the system:

$$\sum_{m=1}^3 \frac{r_{j,-}^{(k)}}{r_{j,-}^{(k)} + \gamma_{m+3}(q)} B_m^{(k)} = -K_{j,-}^{(k)} \left[\widehat{U}_{\text{part}}(q, \cdot; h_0) \right](-h_0).$$

Applying (15) to $\widehat{U}_{\text{part}}(q, z; h_0) = \sum_j \frac{c_{k,j}}{q - \psi_Z^{(k)}(j)} e^{jz}$ yields the explicit entries of $\mathbf{b}^{(k)}$ stated above.

Step 5: matrix form and solution. The three conditions assemble into $\mathbf{M}_{\text{bar}}^{(k)}(q) \mathbf{B}^{(k)} = -\mathbf{b}^{(k)}$. By Lemma 5.4, $\mathbf{M}_{\text{bar}}^{(k)}(q)$ is invertible, so

$$\mathbf{B}^{(k)}(q; h_0) = -(\mathbf{M}_{\text{bar}}^{(k)}(q))^{-1} \mathbf{b}^{(k)}(q; h_0). \quad \square$$

Proof of Lemma 5.4. Subtracting row 1 from rows 2 and 3 replaces entry (i, m) for $i = 2, 3$ by $-\gamma_{m+3}(q)/(r_{i-1,-}^{(k)} + \gamma_{m+3}(q))$. Pulling out -1 from each of these two rows (net sign $(-1)^2 = 1$) and then factoring $\gamma_{m+3}(q)$ from each column gives

$$\det(\mathbf{M}_{\text{bar}}^{(k)}) = \gamma_4 \gamma_5 \gamma_6 \det\left(\frac{1}{x_i + \gamma_{j+3}(q)}\right)_{1 \leq i, j \leq 3},$$

where $x_1 := 0$, $x_2 := r_{1,-}^{(k)}$, $x_3 := r_{2,-}^{(k)}$. The right-hand factor is a Cauchy determinant; the Cauchy formula yields

$$\det\left(\frac{1}{x_i + \gamma_{j+3}(q)}\right) = \frac{\prod_{1 \leq i < i' \leq 3} (x_{i'} - x_i) \cdot \prod_{4 \leq m < m' \leq 6} (\gamma_{m'} - \gamma_m)}{\prod_{i=1}^3 \prod_{m=4}^6 (x_i + \gamma_m)}.$$

Since $x_1 = 0$, the factor $\prod_{m=4}^6 (x_1 + \gamma_m) = \gamma_4 \gamma_5 \gamma_6$ cancels with the prefactor, yielding a ratio whose every factor is nonzero: $r_{1,-}^{(k)} > 0$ since $\eta_{1,-} > 0$; $r_{2,-}^{(k)} > 0$ since $k\eta_{2,+} < 1$; $(r_{2,-}^{(k)} - r_{1,-}^{(k)}) \neq 0$ by (22); the γ_m are pairwise distinct by Assumption 5.1(ii); and $r_{j,-}^{(k)} + \gamma_m \neq 0$ because $-r_{j,-}^{(k)}$ is a pole of $\psi_Z^{(k)}$, so $\psi_Z^{(k)}(-r_{j,-}^{(k)}) = \infty \neq q$, hence $\gamma_m \neq -r_{j,-}^{(k)}$. $\square$

Proof of Proposition 5.5. Step 1: resolvent equation. Taking the Laplace transform of the PIDE $\partial_t p_{\text{surv}} = \mathcal{L}^{(0)} p_{\text{surv}}$ with initial condition $p_{\text{surv}}(0, z; h_0) = 1$ and killing condition $p_{\text{surv}} = 0$ for $z \leq -h_0$, exactly as in Section 5 with $g_k \equiv 1$, gives

$$(q - \mathcal{L}^{(0)}) \widehat{U}_{\text{surv}}(q, z; h_0) = 1, \quad z > -h_0.$$

Step 2: particular solution. The constant forcing $f \equiv 1$ is an eigenfunction of $\mathcal{L}^{(0)}$ with eigenvalue $\psi_Z^{(0)}(0) = 0$, so the particular solution is

$$\widehat{U}_{\text{surv,part}}(q, z) = \frac{1}{q - \psi_Z^{(0)}(0)} = \frac{1}{q}.$$

Step 3: homogeneous solution. By Lemma 5.2 applied with $k = 0$, the admissible homogeneous solution on $z > -h_0$ is

$$\widehat{U}_{\text{surv,hom}}(q, z; h_0) = C_1 e^{\gamma_4^{(0)}(q)(z+h_0)} + C_2 e^{\gamma_5^{(0)}(q)(z+h_0)} + C_3 e^{\gamma_6^{(0)}(q)(z+h_0)},$$

where $\gamma_4^{(0)}, \gamma_5^{(0)}, \gamma_6^{(0)}$ are the roots of $\psi_Z^{(0)}(\gamma) = q$ with $\text{Re}(\gamma) < 0$.

Step 4: boundary conditions. The killing condition imposes, at $z = -h_0^+$:

$$(i) \quad \widehat{U}_{\text{surv}}(q, -h_0^+; h_0) = 0;$$

$$(ii) \ K_{j,-}^{(0)}[\widehat{U}_{\text{surv}}](-h_0^+) = 0, \quad j = 1, 2,$$

by the same argument as in Proposition 5.3: downward jumps from $-h_0^+$ land in the killed region.

Step 5: linear system. Substituting the full solution $\widehat{U}_{\text{surv}} = 1/q + \widehat{U}_{\text{surv,hom}}$ into each condition and applying (15) to the homogeneous terms yields

$$\mathbf{M}_{\text{surv}}(q) \mathbf{C}(q; h_0) = - \begin{pmatrix} 1/q \\ 1/q \\ 1/q \end{pmatrix},$$

where the right-hand side entries are $K_{j,-}^{(0)}[1/q](-h_0^+) = 1/q$ (since $K_{j,-}^{(0)}$ integrates to one against the exponential density). By Lemma 5.6, $\mathbf{M}_{\text{surv}}(q)$ is invertible, so the coefficients $\mathbf{C}$ are uniquely determined.

Step 6: inversion and liquidation probability. Evaluating at $Z_0 = 0$ and inverting via the Bromwich formula gives $p_{\text{surv}}(T, 0; h_0)$. The liquidation probability follows from the complement. $\square$

Proof of Lemma 5.6. The argument is identical to that of Lemma 5.4 with $k = 0$. The phase rates are $r_{1,-}^{(0)} = 1/\eta_{1,-}$ and $r_{2,-}^{(0)} = 1/\eta_{2,+}$, so condition (24) is equivalent to $\eta_{1,-} \neq \eta_{2,+}$. Each denominator factor is nonzero because $-r_{j,-}^{(0)}$ is a pole of $\psi_Z^{(0)}$, hence cannot equal any root $\gamma_m^{(0)}(q)$ of $\psi_Z^{(0)}(\cdot) = q$ for finite q . $\square$

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